

FRONTIER AI SAFETY FRAMEWORK



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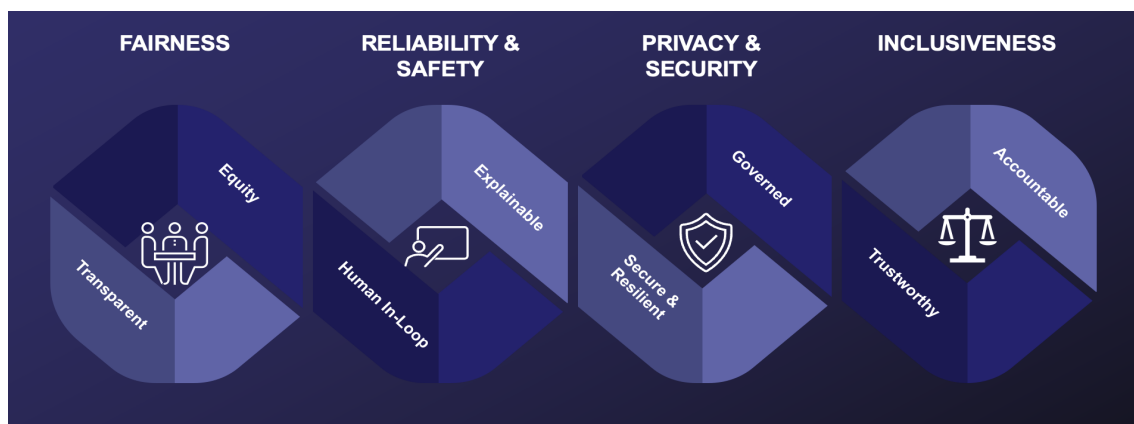
1 Introduction

G42's Frontier AI Safety Framework (“**Framework**”) is a comprehensive set of protocols designed to safeguard G42's advanced AI initiatives by focusing on responsible development, deployment, and oversight of AI technologies. This Framework is tailored specifically to address the unique challenges and risks associated with high-capability AI, building on industry best practices and aligning with established responsible AI principles.

This Framework emphasizes proactive risk identification and mitigation, centering on capability monitoring, robust governance, and multi-layered safeguards to ensure powerful AI models are both innovative and safe. With a systematic approach to early threat detection and risk management, it aims to support G42 in unlocking the benefits of frontier AI safely and ethically. The Framework was developed with input from SaferAI and METR, civil society organizations with expertise in AI risk management frameworks.

Structured as an evolving model with phased implementation, the Framework will adapt to emerging insights and advancements, enabling G42 to implement a resilient and forward-looking AI safety strategy. By setting rigorous standards for security, autonomy, and ethical consideration across operational domains, G42 is equipped to harness transformative AI capabilities while prioritizing safety, security, and public trust.

As a leader in AI innovation, G42 is committed to developing AI systems that align with its principles that prioritize **fairness, reliability, safety, privacy, security and inclusiveness** to reflect and uphold societal values. These principles are displayed in the image below. This Framework ensures that G42's frontier AI models - those with advanced capabilities that may push industry boundaries - are managed with a robust set of safeguards and risk controls that evolve with the models' capabilities.



This approach reinforces G42's commitment to responsible AI practices, ensuring that each AI model and AI system operates safely and securely within the domain of its application (such as medical, energy, investment, language specific).



2 Frontier Capability Thresholds

Capability thresholds establish points at which an AI model's functionality requires substantially enhanced safeguards to account for unique risks associated with high-stakes capabilities. An initial list of potentially hazardous AI capabilities which G42 will monitor for is:

- **Biological Threats:** When an AI's capabilities could facilitate biological security threats, necessitating strict, proactive measures.
- **Offensive Cybersecurity:** When an AI's capabilities could facilitate cybersecurity threats, necessitating strict, proactive measures.

To produce this list, G42 both conducted our own internal risk analysis and received input from external AI safety experts. Initially G42 identified potential capabilities across several domains, including biological risks, cybersecurity, and autonomous operations in specialized fields. We then collaborated with METR and SaferAI to refine our list, prioritizing capabilities based on their potential impact and how feasibly they can be measured and monitored.

In the future, we will map out other hazardous capabilities to consider monitoring. We may also add thresholds for:

- **Autonomous Operation:** When an AI system can make unsupervised decisions with critical implications, particularly in sectors such as healthcare or defense.
- **Advanced Manipulation:** Applicable when AI systems can influence human behavior or decisions on a large scale, warranting enhanced monitoring and usage restrictions.

We plan to integrate decisions on whether to expand our monitoring to include additional hazardous capabilities into our regular framework review process. This includes both our scheduled internal reviews and our annual reviews by third parties. In making these decisions, we expect to consider factors such as: "near miss" incidents, whether internal or industry-wide; recommendations from trusted external experts; as well as changes in industry standards for AI risk management.

G42 will conduct evaluations throughout the model lifecycle to assess whether our models are approaching Frontier Capability Thresholds. At the outset, G42 will conduct preliminary evaluations based on open-source and in-house benchmarks relevant to a hazardous capability. If a given G42 model achieves lower performance on relevant open-source benchmarks than a model produced by an outside organization that has been evaluated to be definitively below the capability threshold, then such G42 model will be presumed to be below the capability threshold.

If the preliminary evaluations cannot rule out proficiency in hazardous capabilities, then we will conduct in-depth evaluations that study the capability in more detail to assess whether the Frontier Capability Threshold has been met. Such evaluations will incorporate capability elicitation – techniques such as prompt engineering, fine-tuning,



and agentic tool usage – to optimize performance, overcome model refusals, and avoid underestimating model capabilities. Models created to generate output in a specific language, such as Arabic or Hindi, may be tested in those languages.

When G42 fine-tunes or adapts an existing open-source model, our assessments of whether our models meet Frontier Capability Thresholds may include relevant evaluation results for the original open-source models when analyzing the impact of our modifications. If we determine that the open-source model has already undergone high-quality assessments for biological threats and offensive cybersecurity capabilities, and our modifications do not enhance its capabilities in these risk areas, we may place greater reliance on the original evaluation results.

In addition to conducting pre-deployment evaluations, post-deployment monitoring will also be used to indicate whether G42's models have reached capability thresholds and whether increased deployment mitigation and security mitigation levels are required.

G42 will publish internal reports providing detailed results of our capability evaluations. These reports will be created for our most advanced models at least once every six months, and the results will be shared with the Frontier AI Governance Board and the G42 Executive Leadership Committee.

G42's Frontier Capability Thresholds are defined in the following table. Each capability threshold is associated with a required Deployment Mitigation Level (DML) and Security Mitigation Levels (SML), which must be achieved before the capability threshold is reached. If a necessary Deployment Mitigation Level cannot be achieved, then the model's deployment must be restricted; if a necessary Security Mitigation Level cannot be achieved, then further capabilities development of the model must be paused. Additionally, if a Frontier Capability Threshold has been reached, G42 will update this Framework to define a more advanced threshold that requires increased deployment (e.g., DML 3) and security mitigations (e.g., SML 3).

Frontier Capability Threshold	Evaluations	Security and Mitigations
Biological threats: Enabling an individual with only introductory biology experience in developing a biological weapon, through providing detailed advice, automating biological processes, etc. Or, facilitating the design of novel biological weapons with substantially greater potential for damage.	<u>Preliminary evaluations:</u> Benchmarking AI models trained on Bio-Chem data (DNA, Proteins, Chemical molecules, etc.) against emerging research to check capabilities to predict and simulate complex biological interactions. Assess language models' accuracy on open-source	DML 2 SML 2 Update the policy to include more advanced thresholds Establish and open source new in-house benchmarks to

	biology benchmarks (e.g., LAB-Bench, WMDP-Bio). <u>In-depth evaluations:</u> Assess Bio-Chem AI models' capabilities to help design harmful substances like new variants. Evaluate language models' ability to answer questions about biological weapons development, relative to expert ability. More ambitiously, evaluate human participants' ability to conduct realistic but safe wet lab experiments with and without language model assistance. Although there is less prior research on evaluating and mitigating risks from biological design tools (BDTs), it is still critical for G42 to develop innovative measures for these tools over time.	evaluate Bio-Chem capabilities.
Offensive cybersecurity: Automating powerful cyber offensive operations against unsecured or secured targets, in a way that could cause critical damage.	<u>Preliminary evaluations:</u> Assess language model performance on open-source cyber offense benchmarks (e.g., Cybench, eyeballvul). <u>In-depth evaluations:</u> Assess model ability to exploit examples of secured targets in a safe	DML 2 SML 2 Update the policy to include more advanced thresholds



	environment.	
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3 Deployment Mitigation Levels

G42's Deployment Mitigation Levels are a set of levels, mapped to the Frontier Capability Thresholds, that describe escalating mitigation measures for products deployed externally. These protect against misuse, including through jailbreaking, as models reach higher levels of capability and risk. These measures address specifically the goal of denying bad actors access to dangerous capabilities *under the terms of intended deployment for our models*, i.e. presuming that our development environment's information security has not been violated.

Deployment Mitigation Level	Description	Specific Measures
Level 1	Foundational safeguards, applied to models with minimal hazardous capabilities. <u>Objective:</u> No novel mitigations required on the basis of catastrophically dangerous capabilities	Examples of foundational safeguards that may be applied include: Model Cards: Documents published alongside each new model deployment, summarising the model's intended use cases, performance on public benchmarks, and the responsible practices conducted to ensure safety. Incident Reporting Channels: Designated pathways for users to report instances of concerning or harmful behavior in violation of company policy to relevant G42 personnel. Information Security Training: Training programs for new and existing personnel on best practices in information security consistent with the measures described in the <i>Security Mitigation Levels</i>.
Level 2	Intermediate safeguards for models with capabilities requiring focused monitoring.	Risk of model misuse is mitigated by: Real-Time Monitoring and Prompt Filtering: Real-time classifiers evaluate inputs and outputs, detecting

	<u>Objective:</u> Even a determined actor should not be able to reliably elicit CBRN weapons advice or use the model to automate powerful cyberattacks including malware generation as well as misinformation campaigns, fraud material, illicit video/text/image generation via jailbreak techniques overriding the internal guardrails and supplemental security products.	and filtering harmful interactions as they occur. This will also be aligned to underlying customer company policy and regulatory compliance. Model Robustness Testing: Regular tests of AI models for robustness against attempts to manipulate or corrupt their output. Asynchronous Monitoring: This off-cycle review catches anomalies missed in real-time, assessing all stored interactions for unusual behaviors. Controlled Rollout: For new frontier-level models, implement phased rollouts, starting with limited access to trusted users, with full deployment only after exhaustive risk assessments.
Level 3	Advanced safeguards for models approaching significant capability thresholds. <u>Objective:</u> Deployment safety should be strong enough to resist sophisticated attempts to jailbreak or otherwise misuse the model.	Risk of model misuse is mitigated by: Real-time anomaly detection and encrypted data handling. Simulation and Adversarial Testing: Regular simulations identify model vulnerabilities and develop adaptive responses. Red teaming activity to identify and mitigate potential risks in the system. Testing is designed to ensure effectiveness across all planned deployment contexts, with specialized subject matter experts providing domain-specific input as needed. Controlled Rollout: For new frontier-level models, implement phased



		rollouts, starting with limited access to trusted users, with full deployment only after exhaustive risk assessments.
Level 4	Maximum safeguards, designed for high-stakes frontier models with critical functions. <u>Objective:</u> Deployment safety should be strong enough to resist even concerted attempts, with support from state programs, to jailbreak or otherwise misuse the model.	To be defined when models reach capabilities necessitating Level 3 deployment mitigation measures

4 Security Mitigation Levels

G42's Security Mitigation Levels are a set of levels, mapped to the Frontier Capability Thresholds, describing escalating information security measures. These protect against the theft of model weights, model inversion, and sensitive data, as models reach higher levels of capability and risk. Each tier customizes protections based on the assessed risk and capability of the model, ensuring G42's AI development remains both resilient and efficient, minimizing disruptions to functionality while maintaining robust security.

Security Level	Description	Specific Measures
Level 1	Suitable for models with minimal hazardous capabilities. <u>Objective:</u> No novel mitigations required on the basis of	None. G42 may choose to open source models.

	catastrophically dangerous capabilities.	
Level 2	Intermediate safeguards for models with capabilities requiring controlled access, providing an extra layer of caution. <u>Objective:</u> The model should be secured such that it would be highly unlikely that a malicious individual or organization (state sponsored, organized crime, terrorist, etc.) could obtain the model weights or access sensitive data.	Access controls and role-based permissions. Model weights are gated by granular role-based permission levels, model access is geofenced to pre-approved locations, limited access attempts using the same credentials. Network segmentation to isolate systems containing model weights. Internal and External Red-Teaming: Rigorous testing by internal security teams, supplemented by external experts, to identify weaknesses. Dynamic Threat Simulation and Response Testing: Regular adversarial simulations expose potential security weaknesses.
Level 3	Advanced safeguards for models approaching hazardous capabilities that could uplift state programs. <u>Objective:</u> Model weight security should be strong enough to resist even concerted attempts, with support from state programs, to steal model weights or key algorithmic secrets.	Model weights and sensitive data are secured through thorough Security Level 2 protocols, as well as the following measures to ensure access to model weights is highly restricted: multi-party and quorum-based approval for high-sensitivity operations, end-to-end encryption of model weights both at rest and in transit, automatic encryption key rotations at regular intervals. Only trusted users with verified credentials are granted access to high-risk models. Access is role-based, aligned with user



		responsibilities, and supported by a zero-trust architecture to prevent unauthorized entry.
Level 4	Maximum safeguards. <u>Objective:</u> Security strong enough to resist concerted attempts with support from state programs to steal model weights.	To be defined when models reach capabilities necessitating Level 3 containment migration measures.

5 Governance and Compliance

A dedicated Frontier AI Governance Board, composed of our Chief Responsible AI Officer, Head of Responsible AI, Head of Technology Risk, and General Counsel, shall oversee all frontier model operations, reviewing safety protocols, risk assessments, and escalation decisions.

Responsibilities of the Frontier AI Governance Board include, but are not limited to:

- i. Framework Oversight: Ensuring this Framework is designed appropriately and implemented effectively, and proposing updates as required.
- ii. Evaluating Model Compliance: Routine reviews of G42's most capable models to ensure compliance with this Framework.
- iii. Investigation: Investigating reports of potential non-compliance and escalating material incidences of non-compliance to the G42 Executive Leadership Committee.
- iv. Incidence Response: Developing a comprehensive incident response plan that outlines the steps to be taken in the event of non-compliance. Incident detection should leverage automated mechanisms and human review, and non-sensitive incident information should be shared with applicable government bodies. We plan for our response protocols to focus on rapid remediation to minimize unintended harmful outputs from models. Depending on the nature and severity of the incident, this might involve implementing immediate containment measures restricting access to the model either externally, internally or both.

G42 will immediately investigate any identified or reported potential instances of non-compliance, taking proportional corrective action as considered necessary.

We will maintain detailed documentation for G42's most capable models, including design decisions, testing results, risk assessments, and incident reports.



Transparency and Reporting

Internal Transparency and Reporting

G42's transparency protocols encourage employees to report concerns while keeping stakeholders informed. These measures include:

- i. **Reporting Mechanisms:** To foster a proactive safety culture, clearly defined channels for reporting security incidents and compliance issues will be established. This includes creating mechanisms for employees to anonymously report potential concerns of non-compliance and ensuring that these reports are promptly addressed.
- ii. **Annual Governance Audits:** G42 will have independent internal audits to verify compliance with our policy.

External Audits and Public Transparency

- i. **External Audits:** To reinforce accountability, G42 will engage in annual external audits to verify compliance with the Framework.
- ii. **Public Disclosure:** G42 will publish non-sensitive, up-to-date and active copies of the Framework. We will share more detailed information with the UAE Government and relevant policy stakeholders. G42 will publish an annual transparency report detailing its approach to frontier models, sharing key insights and fostering public trust.
- iii. **Third - Party Experts:** As deemed appropriate, we will solicit external expert advice for capability and safeguards assessments. This may include partnering with private or civil society organisations with expertise in AI risk management to provide input on our assessments plans and/or internal capability reports ahead of deployment decisions.
- iv. **Threat Intelligence and Information Sharing:** G42 will share threat intelligence with industry partners to address common challenges and emerging risks. G42 will actively participate in forums to set industry standards and share best practices for frontier model safety.

Policy Review

An annual external review of the Framework will be conducted to ensure adequacy, continuously benchmarking G42's practices against industry standards. G42 will conduct more frequent internal reviews, particularly in accordance with evolving standards and instances of enhanced model capabilities .

G42 will proactively engage with government agencies, academic institutions, and other regulatory bodies to help shape emerging standards for frontier AI safety, aligning G42's practices with evolving global frameworks.

Changes to this Framework will be proposed by the Frontier AI Governance Board and approved by the G42 Executive Leadership Committee.



6 Phased Implementation Plan

Phase 1: Framework Initialization (First 6 Months)

- Define capability thresholds and initial safety levels across G42's entities.
- Establish a Frontier AI Governance Board and a Head of Responsible AI.
- Implement foundational security protocols, including real-time monitoring and role-based access controls.

Phase 2: Operational Rollout (6–12 Months)

- Scale up red-teaming exercises, adaptive monitoring, and real-time anomaly detection for high-risk models.
- Conduct routine capability assessments.

Phase 3: Continuous Improvement and Expansion (12+ Months)

- Expand cross-industry collaboration and participate in public safety initiatives.
- Perform annual reviews to adjust Framework elements based on emerging frontier model risks.
- Integrate predictive and adaptive safety technologies to optimize real-time threat response and risk containment.
- Publish an annual transparency report to foster public trust and establish G42's role as a responsible AI leader.

7 Conclusion

G42's Frontier AI Safety Framework establishes the organization as a pioneering force in responsible AI, setting a standard for balancing cutting-edge innovation with comprehensive risk management. By embedding adaptive safeguards, G42 creates a scalable, flexible approach to address the evolving risks associated with powerful AI capabilities. Cross-entity governance ensures that safety protocols are aligned and enforced consistently across the organization, fostering a unified culture of responsibility and ethical AI practices.

Additionally, G42's commitment to external collaboration opens channels for valuable insights, best practices, and oversight from industry experts, academic institutions, and regulatory bodies. This collaborative approach enhances the robustness of the Framework and provides transparency and accountability to the broader AI community.

Through this Framework, G42 demonstrates that it is not only advancing the frontier of AI technology but also leading the industry in safety standards. By ensuring that its frontier AI models deliver meaningful societal benefits, G42 upholds the highest standards of safety, transparency, and accountability—serving as a model for the responsible development and deployment of advanced AI. This proactive stance



positions G42 to continue innovating with confidence while protecting public trust and promoting AI's positive impact on society.

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